

Digital Logic and Design -Lab (EE-172L/CS-130L/CE-222L)

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Topic: Lab 2: BASIC LOGIC GATES

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2.1 Objective

After performing this lab, students should be able to:

- Measure voltage using Digital Multimeter (DMM).
- Verify the behavior of logic gates using the Truth Table (AND, OR, NOT and NOR gates) on breadboard using DMM
- Utilize Datasheet to get familiarized with digital integrated circuits and the internal IC gate level structure.



Figure 1: Objectives

Equipment for today's lab

Exploring Components

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-
1. Logic Circuits are basically electric circuits which implement certain logical function. Before we div *Adjustable Direct Current (DC) Power supply*.
 2. *Digital Multi Meter (DMM)*.
 3. *IC type 7408 AND gate*.
 4. *Light Emitting Diode (LED)*
 5. *Resistors (330 Ohm)*
 6. *Slide switches*.
 7. *IC type 7404 NOT gate*.
 8. *IC type 7402 NOR gate*.

What is an IC?

What is an IC?

- An IC (Integrated Circuit) is a small semiconductor device that contains a collection of electronic components like transistors, diodes, resistors, and capacitors, all integrated into a single chip.
- These components are connected to perform specific functions, such as amplification, logic operations, or signal processing.
- **Why do we need to power up the IC from external voltage source?**

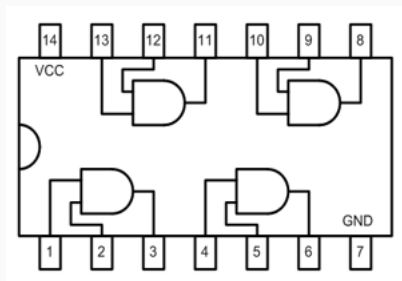


Figure 2: Internal Model of an AND Gate IC

Okay, now you know what is an IC? Can you tell how can we know how many volts to apply to turn on the IC?

What resource should you
refer to?

Task A



Task a

In this task you are required to read the datasheet and get familiarized with internal gate level structure of following ICs.

1. AND gate IC 7408
2. NOT gate IC 7404
3. NOR gate IC 7402

Figure 3: Task A

Logic Gates

What are logic gates?

- Logic gates are the basic building blocks of digital electronics.
- They are electronic devices that perform logical operations on one or more binary inputs to produce a single binary output.
- Logic gates are used to implement Boolean logic, the mathematical foundation for digital circuits.
- Each gate follows a specific logical rule and produces an output based on the combination of inputs.

Different logic gates

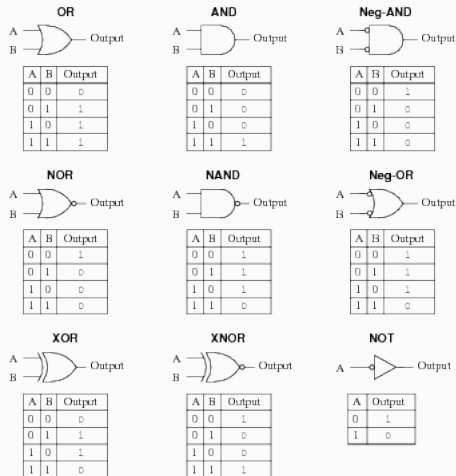


Figure 4: Logic Gates

Task B

AND Logic Gate

i. Design Schematic

1. Figure 2.1 shows the internal diagram of 7408 AND gate taken from datasheet.
2. Figure 2.2 shows the schematic diagram for this experiment.
3. The truth table for an AND gate with two inputs is given in Table 2.1.

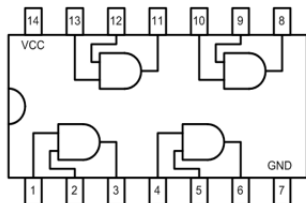


Figure 2.1 - 7408 AND logic gate IC

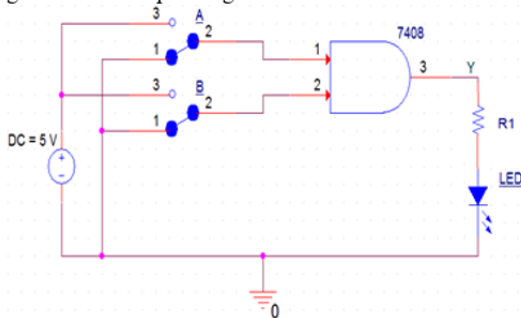


Figure 2.2 - Design schematic of a two-input AND gate.

Figure 5: AND Logic

Task C

NOT Logic Gate

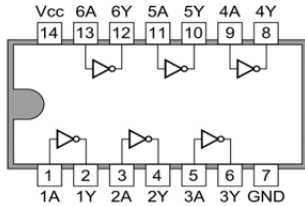


Figure 2.4 - 7404 NOT logic gate IC.

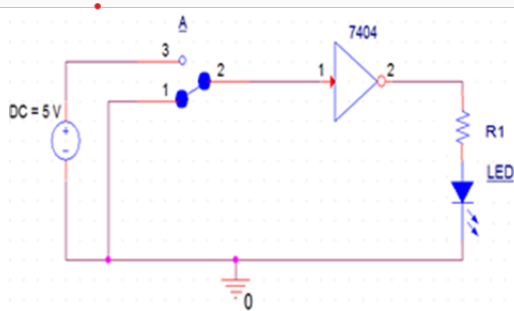


Figure 2.5 - Schematic of NOT gate

Figure 6: NOT Logic

Task D

NOR Logic Gate

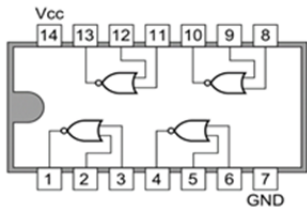


Figure 2.6 - 7402 NOR logic gate IC.

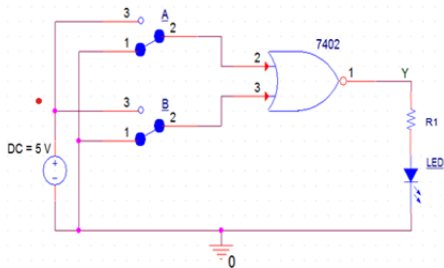


Figure 2.7 - Schematic of NOR gate

Figure 7: NOR Logic

How can you create three
input AND Logic?

How can you create three
input NOR Logic?

Sart your labs! Good luck!
